



# Seafloor debris detection using underwater images and deep learning-driven image restoration: A case study from Koh Tao, Thailand

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## ARTICLE INFO

### Keywords:

Deep learning  
Object detection and classification  
Pre-training strategy  
Seafloor debris  
Super-resolution reconstruction  
YOLOv8

## ABSTRACT

Traditional detection and monitoring of seafloor debris present considerable challenges due to the high costs associated with underwater imaging devices and the complex environmental conditions in marine ecosystems. In response to these challenges, this field study conducted in Koh Tao, Thailand, proposed an innovative and cost-effective approach that leverages super-resolution reconstruction (SRR) technology in conjunction with an optimized object detection model based on YOLOv8. Super-resolution (SR) images reconstructed by seven SRR models were fed into the proposed Seafloor-Debris-YOLO (SFD-YOLO) model for seafloor debris object detection. RDN model achieved the highest reconstruction results with a signal-to-noise ratio (PSNR) of 41.02 dB and structural similarity (SSIM) of 95.08 % and attained state-of-the-art (SOTA) accuracy in debris detection with a mean Average Precision (mAP) of 91.2 % using RDN-reconstructed images with a magnification factor of 4. Additionally, this study provided an in-depth analysis of the influence of magnification factors within the SRR process, offering a quantitative comparison of images before and after reconstruction, as well as a comparative evaluation across various detection algorithms with a novel pretraining strategy. This approach to underwater survey methods, combined with SRR technology, marks an advancement in the field of seafloor debris monitoring, presenting practical solutions to enhance image quality affected by field conditions and enabling the precise identification of marine debris.

## 1. Introduction

### 1.1. A global threat in marine debris

Marine debris, consisting of persistent solid materials discarded in marine and coastal environments, poses a severe threat to ecosystems, economic activities, and human health (UNEP, 2005, 2021; Canals et al., 2021). The oceans, covering 70 % of the Earth's surface, act as major repositories for this waste, which primarily includes plastics, wood, metal, and other materials (Willis et al., 2017; Maes et al., 2018). This debris not only disrupts commercial fisheries and aquaculture but also endangers marine wildlife through entanglement and ingestion, often leading to fatal injuries (UNEP, 2021; Galgani et al., 2013). Over time, these materials degrade into microplastics, which enter the food chain,

posing further risks to marine and human health (Canals et al., 2021; Galgani et al., 2013, 2015). The widespread impact of seabed debris on natural ecosystems, the economy, and human well-being underscores the urgent need for monitoring programs that inform effective coastal management strategies. This need is further highlighted by the increasing body of marine debris-related legislation, which emphasizes long-term monitoring to assess the efficacy of these interventions.

### 1.2. Limitations of traditional monitoring methods

Traditionally, marine debris monitoring has relied on methods such as trawling, optical mapping, and acoustic mapping (Madricardo et al., 2020; Maes et al., 2018; Gerigny et al., 2019). These methods are crucial for the extensive exploration and continuous observation of large seabed

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areas. However, they come with significant limitations. Trawling, which is dependent on weather conditions and suspended particle concentrations, can cause substantial damage to marine habitats, significantly reducing their structure complexity and biodiversity (Shao et al., 2024; Boerger et al., 2010). Optical mapping, often conducted using remotely operated vehicles (ROVs) or autonomous underwater vehicles (AUVs), is constrained by poor visibility, challenging environmental conditions, high costs, and operational complexity (Donohue et al., 2001; Bauer et al., 2008; Pierdomenico et al., 2019). Additionally, the tethered cables of ROVs can restrict coverage and complicate navigation in complex underwater terrains. Acoustic mapping offers broad coverage but struggles with the identification of certain debris types, such as plastics, due to their low acoustic resolution (Spengler and Costa, 2008; Madricardo et al., 2020). These challenges highlight the pressing need for innovative techniques that overcome the limitations of traditional methods, enabling more effective and non-intrusive marine debris monitoring to better preserve marine ecosystems.

### 1.3. Advances in deep learning for marine debris detection

Recent advances in image processing, particularly through deep Convolutional Neural Networks (CNNs), have notably enhanced marine survey fields such as biodiversity monitoring (Saleh et al., 2023), resource assessment (Yekeen and Balogun, 2020), and water quality prediction (Deng et al., 2021). Deep learning has also been extensively applied to marine debris detection (Xue et al., 2021; Fulton et al., 2019), significantly improving the localization and identification of debris in seabed images. This process involves three key steps: Firstly, CNNs extract robust features from seabed images, even under challenging conditions such as light attenuation and interference from suspended particles. The model is designed to learn resilient feature representations that distinguish between debris and the natural seabed environment. Secondly, based on these extracted features, the model further classifies each object within the images into specific categories, identifying various types of debris, including plastics, metals, glass, etc. Thirdly, object detection employs bounding boxes, shaped like a rectangle or square, to specify the locations and sizes with the confidence score of predicted debris objects. Advanced CNN models such as Faster R-CNN (Ren et al., 2015), SSD (Liu et al., 2016), and YOLO (Redmon et al., 2016) have been effectively utilized for accurate seabed debris detection. Faster R-CNN employs a Region Proposal Network for efficient region suggestion and shares convolutional feature maps for higher detection accuracy. SSD combines high detection speed and accuracy by using multi-scale feature maps and default boxes to detect objects of various sizes in a single network pass, eliminating the need for a separate region proposal step. YOLO, which treats detection as a regression problem, maps directly from image pixels to bounding box coordinates and class probabilities, enabling near real-time performance with high accuracy. The YOLOv8 model (Jocher et al., 2023) achieved SOTA results on the MSCOCO classic object detection dataset in 2023, demonstrating significant improvements in efficiency and adaptability. Given the challenges of the underwater environment, integrating YOLOv8 into seabed debris detection shows potential to achieve remarkable speed and accuracy, reducing the need for extensive human intervention and minimizing errors encountered in traditional methods.

### 1.4. SRR in underwater imaging

The quality of underwater images is often limited by water depth and turbidity, which can reduce the accuracy of seabed debris detection, particularly for debris obscured by sediments (Madricardo et al., 2020). Therefore, high-resolution (HR) images are essential to the good performance of the subsequent debris detection process, making researchers continuously upgrade and select high-performance underwater imaging equipment. However, given that most marine debris are small and requires close-range imaging, this approach

complicates data collection, increases reliance on manual labor, escalates costs, and necessitates more advanced and expensive equipment for deeper underwater applications. SRR technology serves as a cost-effective alternative to expensive hardware. SRR improves the image quality by enhancing the visibility and identical features of seabed debris, effectively addressing issues such as fuzziness and motion blur in underwater images (Yue et al., 2016; Zhao et al., 2024a). Additionally, it provides an affordable solution for resource-limited conservation communities, enabling high-quality imaging without expensive equipment. Its advantage lies in generating new pixels to enhance details that are poorly represented in low-resolution (LR) images, where the original resolution is insufficient to capture intricate features or textures necessary for accurate detection and monitoring.

In this study, a novel object detection approach leveraging the SRR technique and an enhanced YOLOv8 model was introduced to achieve automatic seabed debris detection. The key contribution of this research can be concluded as follow: (1) A comprehensive seafloor debris dataset for effective model training was constructed through manual photography in the marine areas of Thailand, encompassing original HR images with 11 meticulously annotated debris categories. To improve the effectiveness of research, a public seafloor debris dataset (Hong et al., 2020) was used for pre-training to enhance robustness, adaptability and transferability of model. (2) This research targets single-image super-resolution (SISR) for underwater images. Specifically, the HR images were degraded to LR images through bicubic interpolation, and the HR and LR pairs were used for training different SRR algorithms. The quality of SR images obtained from the trained networks was evaluated using PSNR and SSIM metrics. (3) An enhanced version of YOLOv8 by incorporating four effective modules, specifically tailored for seafloor detection and named SFD-YOLO was applied to identify debris in three different test set: the original HR test set, the test set reconstructed using Bicubic interpolation, and the test sets reconstructed using various deep learning SRR models. Comparative experiments with seven other detectors demonstrated that SFD-YOLO outperforms them, underscoring its superior performance. Overall, this research aims to make a contribution to the field of marine ecology, particularly in advancing seafloor monitoring and conservation efforts, while explicitly focusing on SISR to address underwater imaging challenges.

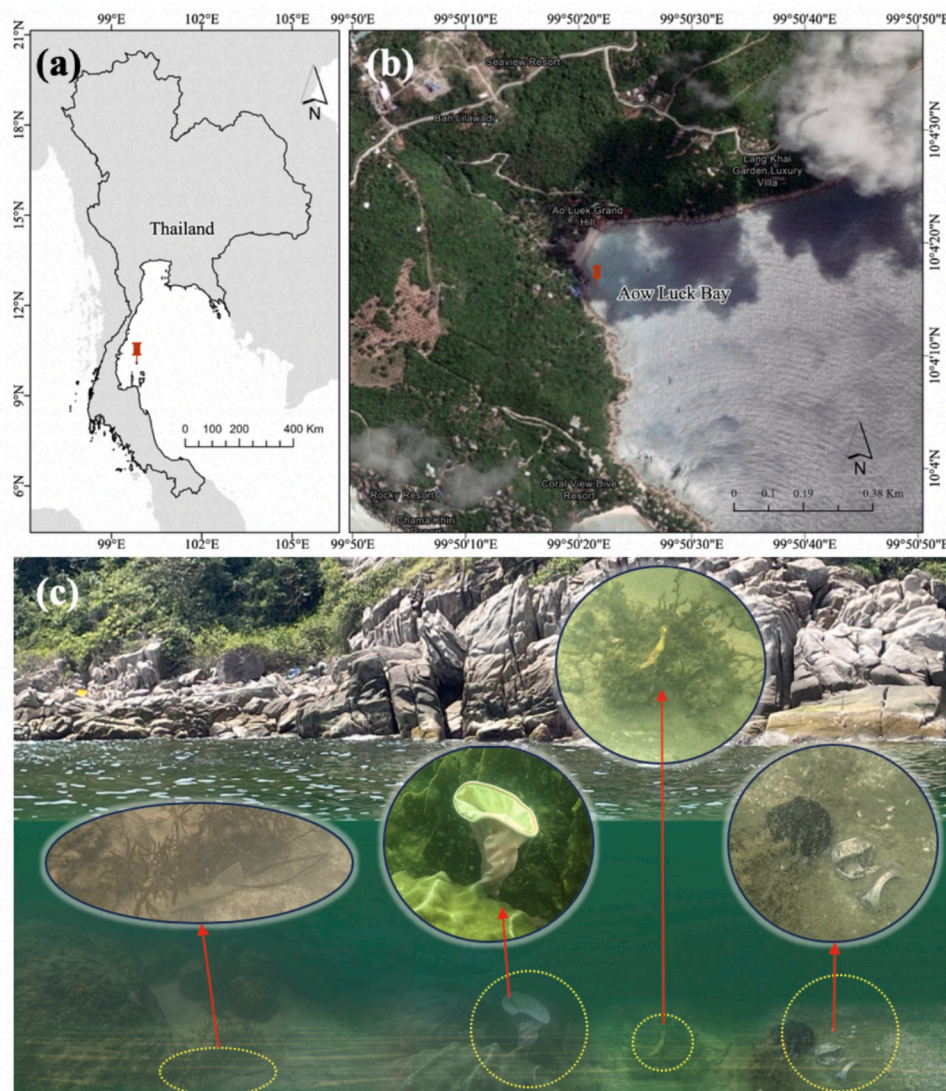
To provide a structured overview of this study, it is organized as follows:

- In Section 2, the study area is introduced, followed by a detailed explanation of the data collection process and dataset preparation. Additionally, the section outlines the SRR techniques employed in this study and presents the development of the proposed SFD-YOLO model for seafloor debris detection.
- Section 3 provides an evaluation of the performance of various SRR models and object detection approaches. This section also includes comparative analyses and ablation studies to investigate the effectiveness of different methods in enhancing detection accuracy.
- A comprehensive analysis is conducted in Section 4, where key factors influencing seafloor debris detection are explored. The impact of magnification factors, the role of transfer learning, and challenges posed by data imbalance are examined in detail. Furthermore, the results obtained in this study are compared with existing research to assess the advancements made.
- Section 5 presents a summary of the key findings and highlights the major contributions of this study. This section also discusses broader implications and outlines potential directions of this study for future research in the field of underwater debris detection.

## 2. Methodology

### 2.1. Study area

Fig. 1 shows the survey was conducted at Aow Luek Bay



**Fig. 1.** (a) Map of survey site at Koh Tao Island, Thailand; (b) Field survey at Aow Luck Bay; (c) Sample of seafloor debris.

(10.0711464 N, 99.8393307E) located on the eastern coast of Koh Tao Island, Thailand, renowned for its stunning marine biodiversity and clear waters, making it a prime spot for snorkeling and diving (Wang et al., 2024; Monchanin et al., 2021; Stuij et al., 2023). The island features vibrant coral reefs, diverse marine life, and sandy seabed, which together create a unique and fragile ecosystem. However, as a popular site for attracting many visitors, the bay is at risk of increased seafloor debris from fishery and human activities, including fishing net, plastic fragments and bottles, leading to pollution and harm to local marine life (Ballesteros et al., 2018). The accumulation of marine debris in the bay poses significant threats to its delicate ecosystem, including damage to coral reefs, entanglement of marine species, and the introduction of pollutants into the water (Koiwanit and Filimonau, 2023). These factors make Aow Luek Bay a strategic location for deploying innovative monitoring tools and applying debris detection algorithms.

## 2.2. Field survey and datasets

An algae bloom during data collection caused poor visibility a few meters below the surface, making towed underwater survey equipment like Aerial-aquatic speedy scanner (AASS) (Zhao et al., 2024a; Zhao et al., 2024c) and Speedy Sea Scanner (SSS) (Ma et al., 2024; Terayama et al., 2022; Mizuno et al., 2019) unsuitable. However, 1–2 m above the

seafloor, visibility was unaffected. Therefore, using professional divers equipped with consumer-grade GoPro cameras were engaged to acquire the high-quality visual data became the only viable survey method. These divers, proficient in navigating the bay's diverse underwater terrains, were responsible for recording continuous video footage, ensuring a thorough survey of the area. The GoPro cameras were chosen for their ability to capture 4 K resolution footage at a consumer-grade cost. This capability was critical for capturing the intricate details of debris and other seafloor features with exceptional clarity. The compact and durable design made them ideal for underwater exploration, offering resistance to water pressure and the harsh saline environment. Additionally, the wide-angle lens provided extensive coverage of the seabed, enabling a broad view of the underwater ecosystem within a single frame.

The video footage was meticulously processed frame by frame, extracting images at a resolution of  $3840 \times 2160$  pixels. Frames without debris were discarded, and the remaining images were manually annotated into 11 debris categories with their respective proportions: plastic bags (21.88 %), bottles (17.19 %), sheets (11.98 %), solid objects (6.25 %), boxes (6.25 %), metal sheets (2.08 %), tubes (30.21 %), glass bottles (1.04 %), fishing nets (0.52 %), ropes (1.04 %), and cans (1.56 %). Each category was labeled with numerical identifiers ranging from 0 to 10, using the LabelImg tool. Due to the increasing number of tourists in Aow

Luek Bay, plastic debris constitutes a significant proportion, leading to a higher volume of improperly discarded plastic debris. Moreover, frequent maritime activities, such as fishing and boating, can attributed discarded or lost metal tube. Since the raw data was limited but had high resolution, the images were cropped into smaller, HR patches of  $512 \times 512$  pixels to increase the dataset size. After splitting the dataset into training and validation subsets with a 9:1 ratio, data augmentation was applied exclusively to the training set, resulting in a final dataset of 294 patch images.

To simulate real-world conditions and improve survey efficiency while reducing operational costs, HR images were subsequently down-sampled using an image degradation model. This process ensured that the dataset included both HR and corresponding LR images, which are essential for effectively training and evaluating SRR models. The degradation model can be expressed as follows (Timofte et al., 2017; Zhang et al., 2021):

$$g = (f \otimes h) \downarrow_s^{\text{bicubic}} + \eta \quad (1)$$

where,  $g$  represents a LR image,  $f$  represents a HR image, and  $h$  denotes the point spread function under the uniform linear motion model. The symbol  $\otimes$  indicates the convolution operation,  $\downarrow$  signifies the down-sampling operation,  $s$  is the magnification factor, bicubic refers to the interpolation algorithm, and  $\eta$  represents the additive Gaussian white noise.

### 2.3. Super-resolution reconstruction

Fig. 2 shows the flowchart of this study. Following the degradation process described in the previous section, data augmentation techniques were applied to both HR and LR images to diversify data and simulate various natural variations, thereby improving the robustness of the SRR models. The augmented images were then used as input for seven SRR algorithms, including the traditional Bicubic method (Khaledyan et al., 2020) and six deep learning-based algorithms with diverse design strategies introduced in the following section, to reconstruct LR images and output high-quality SR images. These SRR models were optimized on the same training and validation set to minimize the difference between the output and the ground truth HR images using the loss functions and evaluation metrics such as PSNR and SSIM. Ultimately, these SRR

models can take new LR images such as those obtained during underwater surveys for marine debris and output SR images that exhibit finer details and higher quality than the original LR images.

#### 2.3.1. Network architectures

Deep learning-based image SR techniques primarily focus on reconstructing HR images from a single LR input. Since LR images inherently contain less information than their HR counterparts, SRR models must infer high-frequency details, requiring sophisticated inference mechanisms capable of accurately interpolating or generating these details. In recent years, researchers have proposed various SRR models based on diverse deep learning network designs, as illustrated in Fig. 3, including: (1) linear networks (Dong et al., 2015), (2) residual learning networks (Lim et al., 2017; Zhang et al., 2018b) (3) dense connection networks (Zhang et al., 2018a), (4) recursive learning networks (Li et al., 2019) and (5) GANs (Wang et al., 2018). In this study, six deep learning-based SRR algorithms were employed to perform pixel reconstruction on underwater LR images to generate SR images and their reconstruction performance was evaluated against the traditional Bicubic interpolation method, which is commonly used as a baseline due to its historical effectiveness (Xiang et al., 2022; Wang et al., 2021a, 2021b).

One of the linear models, SRCNN, introduced by Dong et al. (2015), marked a significant breakthrough in applying deep learning to SRR, transforming it from a simple model to an end-to-end learning process. Unlike traditional methods, SRCNN employs only three convolutional layers:  $f1 \times f1$  patch extraction and representation, nonlinear mapping, and  $f3 \times f3$  reconstruction, achieving significantly higher PSNR values. EDSR, a residual learning network developed by Lim et al. (2017), improved upon SRCNN by removing batch normalization (BN) layers and selecting the L1 loss function. This simplification allowed the network architecture to be deepened and widened, enhancing performance. RDN also a residual learning network introduced by Zhang et al. (2018b), incorporated dense connections and residual dense blocks, enabling the network to utilize features from all layers and allowing for rich feature extraction and accumulation. The combination of local features from individual residual dense blocks (RDBs) with global features, enabling the network to leverage both local and global contextual information when reconstructing images. RCAN, developed by Zhang

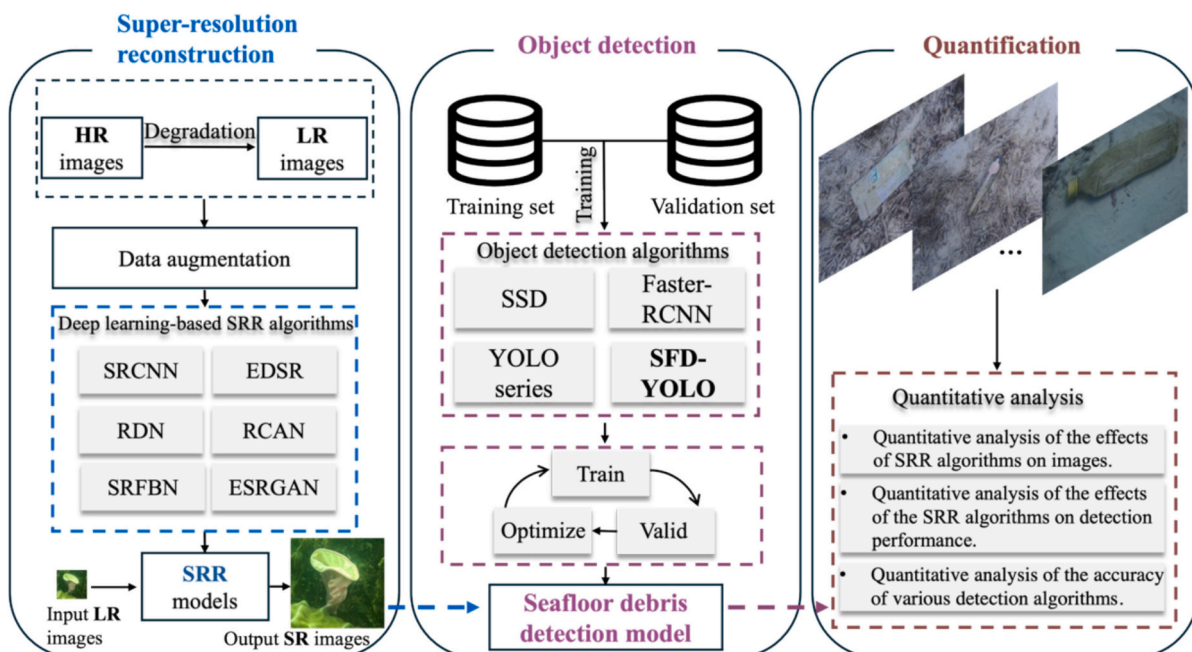


Fig. 2. Flowchart of automatic seafloor debris detection.

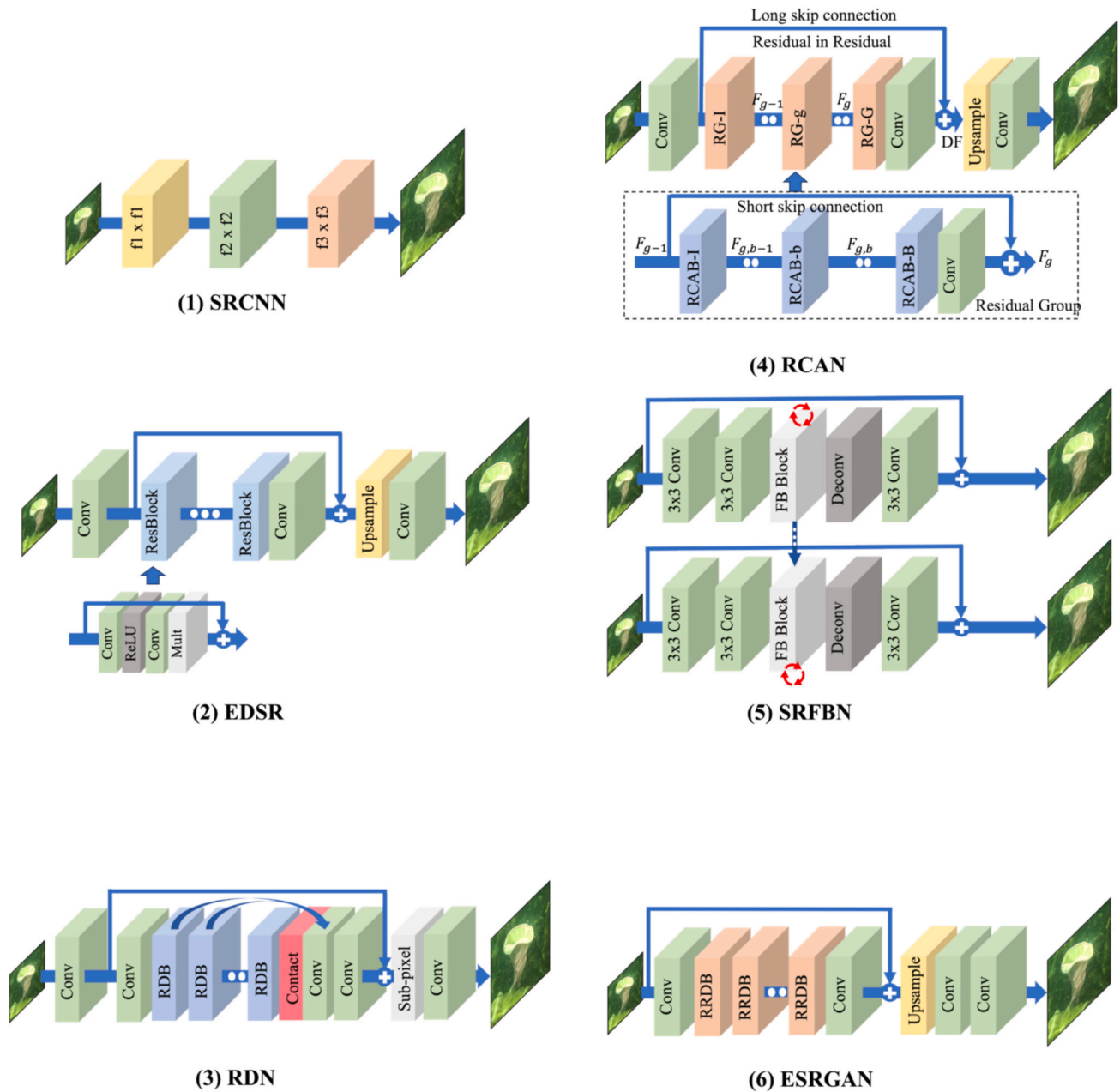


Fig. 3. Architectures of SRR networks.

et al. (2018a), made a significant advancement by integrating channel attention mechanisms (CAM), which adaptively scale features, emphasizing more informative ones. Residual Channel Attention Blocks (RCABs), the core component of RCAN, use CAM to recalibrate channel-wise feature responses by explicitly modeling interdependencies among channels, enabling the reconstruction of finer details and textures in LR images. SRFBN, adopting recursive learning strategies and proposed by Li et al. (2019), introduced feedback mechanisms within a recurrent structure, allowing the network to iteratively refine its output. The network utilizes back-projection techniques to maintain consistency between LR and HR images, resulting in progressively improved image reconstruction quality with each feedback loop. ESRGAN, developed by Wang et al. (2018), implemented the principles of GANs, building upon the Super-Resolution Generative Adversarial Network (SRGAN) with residual-in-residual dense blocks and a relativistic discriminator. This approach generates photorealistic HR images from LR inputs, with the resulting textures and details appearing more natural and finely detailed

to human eyes.

### 2.3.2. Network training and optimization

The training stage of SRR models involves feeding LR images and their corresponding HR images into the models, followed by updating the model parameters through gradient descent and backpropagation. All experiments involving the SRR models were implemented using PyTorch and optimized with the following hyperparameters: an initial learning rate of  $5 \times 10^{-5}$ , with a decay factor of 0.5 applied every 200 epochs. After a total of 300 epochs, with a batch size of 16, the models can fit to the training data.

### 2.3.3. Evaluation metrics

To rigorously evaluate the reconstruction results of SRR algorithms on LR images, this study employed PSNR and SSIM as evaluation metrics (Chen et al., 2022). PSNR is a widely used metric for assessing the quality of reconstructed images relative to their original HR

counterparts. It is particularly common in the fields of image and video compression, as well as in SR. PSNR is defined in terms of the Mean Squared Error (MSE) between the reconstructed image and the original image. A higher PSNR value indicates better image quality. The formula for calculating PSNR is as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (I_{SR}^i - I_{HR}^i)^2 \quad (2)$$

$$PSNR = 10 \cdot \log_{10} \left( \frac{MAX_I^2}{MSE} \right) \quad (3)$$

SSIM is a metric used to compare the similarity of luminance, contrast, and structural details between images. Unlike PSNR, SSIM is more aligned with the human visual system's sensitivity to structural information in images. SSIM values range from 0 to 1, where a higher value indicates greater structural similarity and less distortion, reflecting better image quality. The principle implementation of SSIM is defined as:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + K_1)(2\sigma_{xy} + K_2)}{(\mu_x^2 + \mu_y^2 + K_1)(\sigma_x^2 + \sigma_y^2 + K_2)} \quad (4)$$

where,  $\mu_x$  and  $\mu_y$  represent the average luminance of images  $x$  and  $y$ , while  $\sigma_x$  and  $\sigma_y$  denote the standard deviations of  $x$  and  $y$ .  $\sigma_{xy}$  is the covariance of  $x$  and  $y$ . Constants  $K_1$  and  $K_2$  are introduced to prevent division by zero in the denominator. Combining PSNR and SSIM allows for a balanced evaluation, ensuring that the SR images are not only mathematically accurate but also visually pleasing and structurally consistent with the original HR images.

## 2.4. Seafloor debris detection

### 2.4.1. Network architecture of SFD-YOLO

Seafloor debris presents significant challenges due to the variability in material and appearance, compounded by low light, turbidity, and natural backgrounds that often resemble the debris itself, which severely degrade image quality and reduce detection accuracy. In these complex environments, debris is often partially or fully buried under sand and stones, increasing the risk of false positives and missed detections. While classical YOLOv8 is effective in various scenarios, its adaptability to seafloor conditions with very low-resolution images can be limited when compared to network architectures specifically designed for such conditions. To address these challenges in deep-sea environment research, the proposed SFD-YOLO, as depicted in Fig. 4 (a), utilizes network architectures specifically designed for such conditions.

Integrated innovative structures in SFD-YOLO is depicted in Fig. 4. Firstly, large object detection head that is specifically designed to localize and identify large sized objects like plastic bags, plastic bottles through an expanded receptive field and improved feature extraction was added. Secondly, the pooling pyramid module of original YOLOv8 was replaced with Attention-based Intra-Scale Feature Interaction (AIFI) module (Zhao et al., 2024a, 2024b, 2024c) for the enhancement of multi-scale feature utilization in challenging underwater environments. By integrating attention mechanism in the AIFI module, the model can adjust weights of feature dynamically to cope with the complex and challenging underwater environment. Thirdly, the Omni-Dimensional Dynamic Convolution (OD Conv) (Li et al., 2022) is introduced in the network head to dynamically adapt to various target sizes and shapes. Finally, the attention mechanism module (Yang et al., 2021) was integrated into the last three C2F modules of the backbone network. Emphasizing important features while minimizing irrelevant ones like rocks or seagrass.

In real underwater scenarios, debris often occupies a significant portion of the image, and due to variations in shooting distances, the size of the same object can vary greatly. These issues can lead to an increased

rate of false detections. To improve detection comprehensiveness, large object detection head (LODH) was added to the original head layer of the model to enhance the ability to detect large debris. In this way, even with varying object sizes across different images, the model can recognize them as the same object during training, compensating for detection errors caused by distance and angle variations.

While the SPPF module in original YOLOv8's neck layer enables the network to capture background information with varying levels of detail through feature aggregation at different spatial scales, its frequent pooling operations significantly reduce the spatial resolution of feature maps, leading to the loss of fine details such as the edges and textures of debris. To address these challenges, the replacement of the original SPPF module in YOLOv8 with AIFI module (Zhao et al., 2024a), highlighted in Fig. 4 (c), allows the network to adaptively weight and capture more nuanced contextual information by emphasizing the importance of specific features. This dynamic adjustment strategy ensures the retention and enhancement of the fine details that are often lost with traditional pooling operations, significantly improving overall accuracy and robustness of object detection in complex seafloor environments.

The static convolution operation in original YOLOv8 is typically used to process all input images, while this approach struggles with complex backgrounds, leading to reduced sensitivity to important regions. Although deepening the network can enhance accuracy, this often results in an excessive number of parameters, making model training process time-consuming and resource intensive. Omni-Dimensional Dynamic Convolution (ODConv) (Li et al., 2022) module illustrated in the Fig. 4(b) incorporates attention mechanisms across multiple dimensions, significantly enhancing feature extraction and contextual information capture. Its advantages to adjust convolutional kernels based on the input, allowing the model to adapt to varying image complexities and improving performance in challenging seafloor setting without deepening network architecture.

Fig. 4(a) illustrates the C2f-SimAM module which integrates of SimAM module with C2f in SFD-YOLO. SimAM module, a lightweight 3D attention mechanism, highlighted in the Fig. 4(c), enhances the representational capacity of feature maps through 3D-weights Generation, Expansion and Fusion. Initially, the input data is processed to generate 3D attention weights and the feature map of input data is then expanded with 3D weight through multiplications. Finally, the enhanced features are fused with the original input features, combining the refined and original information to produce a more informative and robust feature representation. In this way, model effectively amplifies critical features while preserving the original information, resulting in a more comprehensive and nuanced representation of the input data.

### 2.4.2. Network training and optimization

The YOLO object detection system is engineered to simultaneously perform classification and localization during its training process. The entire model is trained end-to-end to predict both class probabilities and the center coordinates, width and height of bounding box with utilizations of classification loss and localization loss. In the experiment, the training of YOLO was conducted on an RTX 4090 GPU with 64 GB of RAM. All models were trained using the Adam optimizer to update parameter of model, with a batch size of 32, for 200 epochs, and with a learning rate of  $10^{-4}$ .

### 2.4.3. Evaluation metrics

Four groups of commonly used evaluation metrics were adopted in object detection: Precision, Recall, Intersection-over-Union (IoU) and mAP at 50 % IoU (mAP@0.5) (Géron, 2019; Everingham et al., 2010). The formulas for these evaluation metrics are as follows:

Precision: Measures the proportion of true positive detections out of all positive detections (both true and false).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

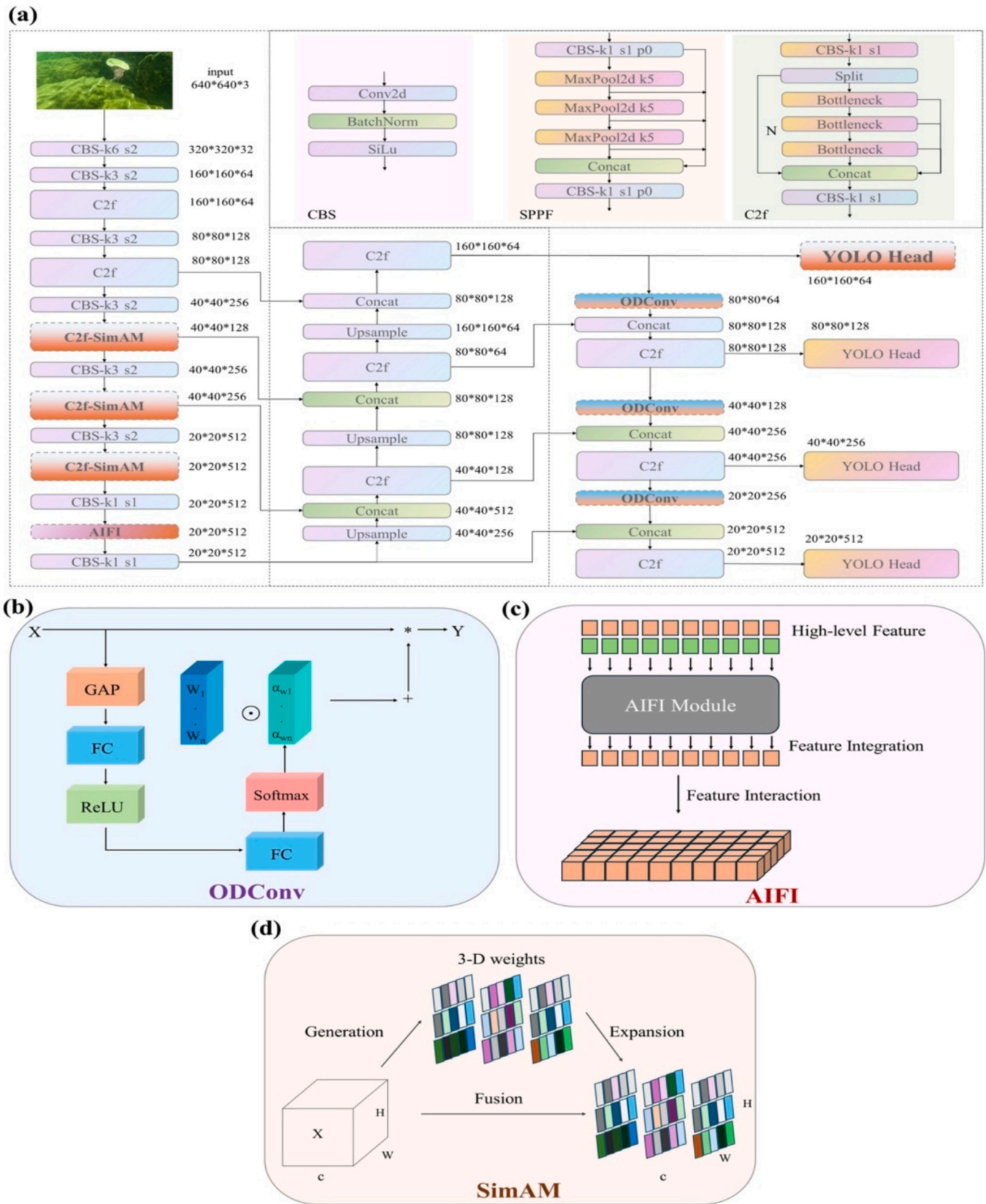


Fig. 4. Structure of proposed SFD-YOLO: (a) overall architecture; (b) ODConv module; (c) SimAM module; (d) AIFI module.

where TP represents true positives, and FP represents false positives. High Precision indicates fewer false positive results.

Recall: Measures the proportion of true positive detections out of all actual positives (true positives and false negatives).

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

where FN represents false negatives. High recall shows that the model correctly identifies a higher number of true positive examples.

IoU: Measures the area of the overlap between the predicted bounding box and the ground truth box by the area of the union of these two boxes.

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (7)$$

mAP@0.5: The mean of the Average Precision (AP) across all classes with IoU threshold set to 0.5. It is computed as:

$$\text{mAP@0.5} = \frac{1}{N} \sum_{i=1}^N \left( \sum_{j=0}^{n-1} (r_i - r_{j+1}) p \right) \quad (8)$$

where N is the number of classes, r denotes Recall, and p denotes Precision. The summation is used to calculate the area under the Precision-Recall curve for each class.

### 3. Experiment results

#### 3.1. Reconstruction results of various SRR models

The performance of seven SRR models was trained on the training and validation set and the generated SR images were then evaluated by using PSNR and SSIM metrics. The reconstruction results on the testing set are shown in Table 1, demonstrating the superiority of CNN-based SRR methods over the traditional Bicubic interpolation method. The RDN network based on the residual dense network achieved the best results with a PSNR of 41.016 dB and SSIM of 95.08 % at x4 magnification factor, outperforming the Bicubic interpolation method which had a PSNR of 38.63 dB and SSIM of 92.53 %. ESRGAN yield the worst performance among the deep learning-based models, even lower than Bicubic interpolation method. The reasons for its poor outcome are that PSNR and SSIM measure pixel-level differences, focusing on the absolute error between the generated and real images. This contrasts with the core design concept of ESRGAN, which seeks to create new visual details that may not exist in the original image (Sara et al., 2019). While these added details can enhance visual effect, they may result in lower PSNR and SSIM scores. Additionally, perceptual loss emphasizes capturing high-level features and structural similarities, rather than ensuring pixel-to-pixel accuracy (Johnson et al., 2016). Consequently, images optimized for perceptual quality might not score as highly on traditional metrics like PSNR and SSIM, despite providing a better visual experience. Notably, while the ESRGAN model did not perform as well on metrics, the reconstructed images appeared more realistic than those produced by other models, effectively reproducing textures absent in the HR images.

Fig. 5 displays SFD-YOLO detection results around ground truth HR images and generated images from different SRR models. The reconstruction capability of the Bicubic interpolation technique is insufficient when applied to underwater imagery, which often contains substantial

noise. As a result, it produces relatively low confidence scores and one missed detection. In contrast, deep learning-based approaches demonstrate superior reconstruction performance, improving the clarity of seafloor debris images and enhancing detail. Among these, RCAN and RDN models achieved higher reconstruction quality, ensuring no missed detections when applied with SFD-YOLO, in comparison to ESRGAN, SRCNN, EDSR, and SRFBN. RDN model demonstrated an optimal performance both on confidence scores of 0.96 and visual effects on the SR images among various SRR models.

#### 3.2. Object detection results of SFD-YOLO

As described in Section 2.4.1, the original YOLOv8 model struggles to adapt to the complexities of the seafloor environment. To enhance the detection of seafloor debris, the SFD-YOLO model was developed by integrating four effective algorithmic modules into backbone, neck and head network. Classical YOLOv8 achieved mAP of 84.5 %, serving as the baseline. The ablation experiments were conducted on the test set by isolating each introduced module as shown in Table 2.

Firstly, a LODH designed to localize and identify large-sized objects was added. The original YOLOv8 model employed three detection heads designed to balance detection performance for small and medium-sized objects. However, the seafloor environment contained a substantial amount of large debris, which the original detection heads struggled to detect effectively. By introducing an additional detection head specifically for large objects and connecting it to deeper feature maps, the model captured information with a larger receptive field. This modification improved the Recall from 76.5 % to 80.6 % and increased the mAP by 2.9 %.

Building upon YOLOv8-LODH as the new baseline, AIFI module was introduced into the backbone network to enhance seabed debris detection performance in complex underwater environments. The AIFI module incorporates advanced Self-Attention Blocks and Cross-Scale Feature Fusion mechanisms, which enabled the network to better discriminate objects that were visually similar to the background by capturing contextual relationships, while also facilitating the effective integration of multi-scale features. This integration specifically addressed challenges such as low visibility, cluttered backgrounds, and varying object scales in underwater environments, resulting in a 2.8 % increase in mAP over the new baseline.

Integrating the ODCnv module into the head of the YOLOv8 network enhanced the model's ability to focus on critical features. This module employed dynamic convolutional kernels that adaptively adjust to input features across spatial and channel dimensions. Since seafloor debris often exhibited significant variations in size, shape, and texture, and may blend with the surrounding environment, the dynamic nature of the kernels enabled the model to adaptively improve attention on relevant features and suppress irrelevant or noisy information. This architectural enhancement led to a further 0.7 % increase in mAP while maintaining computational efficiency.

Lastly, integrating the SimAM module into the network introduced efficient 3D attention mechanism by modeling the importance of each neuron based on the variance of its activation. This approach effectively captures spatial, channel, and contextual information in a parameter-free manner. In the context of seafloor debris detection, this module was particularly suitable for it enhanced the model's ability to discriminate subtle differences between the debris and the surrounding environment. By adaptively adjusting attention weights across spatial and channel dimensions, the module improved the detection of both small

**Table 1**  
Evaluation metrics with x4 magnification factor on LR test set.

| Metrics (%) | Bicubic | ESRGAN | SRCNN | EDSR  | SRFBN | RCAN  | RDN   |
|-------------|---------|--------|-------|-------|-------|-------|-------|
| PSNR (dB)   | 38.63   | 38.21  | 40.11 | 40.84 | 40.84 | 41.00 | 41.02 |
| SSIM (%)    | 92.53   | 91.84  | 94.26 | 94.94 | 94.95 | 95.07 | 95.08 |

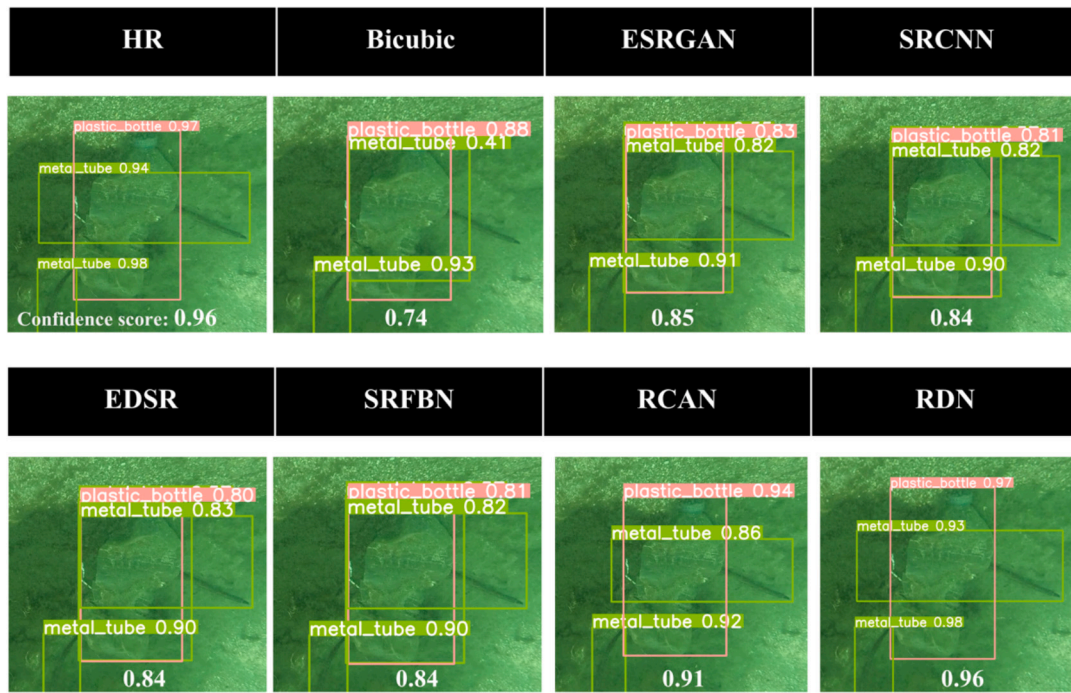


Fig. 5. Comparison of the detection results of seafloor debris images reconstructed using seven different methods.

Table 2

Ablation experiments of SFD-YOLO on different modules. A checkmark (✓) indicates that the module or improvement strategy was utilized.

| Baseline | LODH | AIFI | ODConv | SimAM | P (%) | R (%) | mAP@0.5 (%) |
|----------|------|------|--------|-------|-------|-------|-------------|
| YOLOv8s  | ✓    |      |        |       | 91.0  | 76.5  | 84.5        |
|          | ✓    |      |        |       | 86.2  | 80.6  | 87.4        |
|          | ✓    | ✓    |        |       | 86.1  | 77.4  | 90.2        |
|          | ✓    | ✓    | ✓      |       | 82.4  | 80.3  | 90.9        |
|          | ✓    | ✓    | ✓      | ✓     | 87.5  | 81.2  | 91.2        |

and large objects, which was crucial in underwater environments where object scales vary widely without substantially increasing computational parameters, leading to a further 0.3 % increase in mAP.

Table 3 presents the detection results of the SFD-YOLO model across SR images produced by seven SRR models and ground truth (original) HR images on test sets. SFD-YOLO demonstrates consistent detection performance on SR test set generated by six different deep learning-based SR algorithms, with mAP ranging from 84.3 % to 89.7 %, effectively balancing Precision and Recall as the overall evaluation metric. The detection algorithm achieves its highest mAP of 89.7 % when using the RDN for image reconstruction, showing only minimal deviation of 1.2 % mAP from the results obtained directly from the HR test set. While ESRGAN shows comparatively weaker performance in terms of PSNR and SSIM, the SR images it generates contain rich details, resulting in 84.3 % mAP and notably surpassing the detection results on SR images reconstructed based on Bicubic interpolated methods, which achieved only 80.5 % mAP.

## 4. Discussion

### 4.1. Effect of magnification factor

In SR applications, the magnification factor is a critical parameter, representing the ratio by which a LR image is enlarged to restore it to a HR image. To determine the optimal magnification factor that maximizes the accuracy of seafloor debris detection, it is essential to comprehensively examine the imaging outcomes and enhancement capabilities associated with each magnification factor. The LR images were derived from HR images initially captured at a resolution of  $512 \times 512$  by underwater cameras. Following a  $4\times$  downsampling process, as outlined in Eq. 1, the HR images were downsampled to create x1-LR ( $128 \times 128$ ) images. Similarly, the HR images were downsampled by factors of 2, 3, and 4 to generate x2-LR ( $256 \times 256$ ), x3-LR ( $384 \times 384$ ), and x4-LR ( $512 \times 512$ ) images, respectively.

To explore the optimal SRR magnification factor, the deep learning-based SRR model RDN was selected and trained on these LR images. The LR images and their corresponding scaled HR images were fed into the RDN model to produce four SRR models: SRR-1, SRR-2, SRR-3, and SRR-4. The x1-LR dataset was then input into these models to generate outputs of x2-SR ( $256 \times 256$ ), x3-SR ( $384 \times 384$ ), x4-SR ( $512 \times 512$ ), and

Table 3

Evaluation metrics for detection results on the test set reconstructed by different SRR algorithms.

| Metrics (%) | HR   | Bicubic | ESRGAN | SRCNN | EDSR | SRFBN | RCAN | RDN  |
|-------------|------|---------|--------|-------|------|-------|------|------|
| mAP@0.5     | 91.2 | 80.5    | 84.3   | 86.6  | 89.1 | 88.9  | 89.5 | 89.7 |

x5-SR (640 × 640), which were subsequently used for debris detection tasks with the SFD-YOLO model. The accuracy of object detection at different magnification factors was analyzed, as presented in Table 4. As magnification increased, the gap between the SFD-YOLO detection results on HR images and SR images progressively narrowed. When magnification factor increased from 2× to 3×, the richness of details and feature representation in the SR images was improved, leading to a significant boost in detection accuracy, with mAP rising from 82.5 % to 89.2 %. At a magnification factor of 4×, the gap further narrowed, reducing the difference in mAP between RDN-generated SR images and HR images to just 1.5 %. However, increasing the magnification factor to 5× led to a decline in mAP due to excessive noise amplification, making debris detection more challenging (Zhao et al., 2025; Hu et al., 2018). Overall, SR images reconstructed by the RDN with an optimal magnification factor of 4×, made a good trade-off between reconstruction performance and image noise, effectively enhancing image detail while minimizing noise interference, thereby improving detection accuracy.

Fig. 6 showcases the confidence scores of SFD-YOLO at various magnification factors for seafloor debris detection. These scores reflect the probability, ranging from 0 to 1, that the object detection model assigns to the presence of objects as a specific category within the images. The results indicate that the detection confidence scores of SFD-YOLO on SR images generated by RDN model are relatively significantly higher compared to those obtained with original LR images, underscoring the effectiveness of deep learning-based SRR algorithms. As the magnification factor increases, the SR images become sharper and more detailed, further improving detection accuracy. Among the various magnification factors, the x4 level achieves the highest detection performance, delivering optimal results for detecting seafloor debris.

#### 4.2. Effect different object detection models on SR images

To evaluate the effectiveness of SFD-YOLO in the seafloor debris detection algorithm multiple deep learning-based object detection models, encompassing both single-stage detectors (e.g., YOLO series) and two-stage detectors (e.g., Faster R-CNN, SSD), were trained on x4-SR images generated by the RDN network to assess their performance in seafloor debris detection as shown in Table 5. These eight comparative models are widely utilized in object detection and are increasingly applied in marine debris detection (Xue et al., 2021; Politikos et al., 2021), making them suitable as benchmark models. These models exhibited minimal differences in detection results between SR images generated by the RDN and HR images, with variations ranging from 0.3 % to 3 %, which confirms expectations and highlights the significant advantages of deep learning-based SRR models in enhancing detection accuracy. In contrast, the detection performance of YOLOv5Lite-c, a light-weight version of YOLOv5, was less satisfactory across the three test sets by achieving a 55.4 % of mAP on the Bicubic set, which was unexpectedly higher than on the HR and RDN-generated sets. This result can be attributed to its simplified network structure, which lacks the sensitivity to the finer details and textures added by the RDN. After using from SR images generated by Bicubic interpolation to those generated by RDN, the detection results of YOLOv4 showed a significant improvement, with a 19.5 % increase in mAP. Among them, YOLOv8 achieved a high mAP of 84.5 % and its enhanced version, the innovative SFD-YOLO model further increased the mAP to 91.2 %, marking it a SOTA model in detecting seafloor debris.

**Table 4**  
The results of object detection under different SR magnification factors.

| Metrics (%) | HR   | x1-LR | x2-SR | x3-SR | x4-SR | x5-SR |
|-------------|------|-------|-------|-------|-------|-------|
| mAP@0.5     | 91.2 | 73.9  | 82.5  | 89.2  | 89.7  | 89.1  |

#### 4.3. Effect of transfer learning with the TrashCan dataset

In seafloor debris detection research, acquiring datasets is often cumbersome, difficult, and costly due to the extensive time, manpower, and resources required for image data collection. This process typically involves specialized imaging equipment and operations in challenging environments. Additionally, annotating the collected data demands significant time and effort, requiring meticulous attention and expert knowledge, making it difficult to independently build large, high-quality dataset quickly. The complexity and variability of the underwater environment further complicate data collection efforts. In this research, transfer learning refers to improve the performance of target model on seafloor debris domains by leveraging from related source domains. In essence, the model is first pre-trained on a large open-source dataset to learn general features and distribution with its parameters to fit on this dataset. It then fine-tuned on a smaller dataset, requiring only minor parameter adjustments to adapt to the new distribution of data. In this way, the dependence on a large number of target domain data can be reduced for constructing target model (Zhuang et al., 2020).

Recent research initiatives have resulted in the creation of a comprehensive dataset known as TrashCan (TC) (Hong et al., 2020), which contains a substantial collection of debris images, each meticulously labeled with corresponding annotations. This dataset significantly reduces the time and cost of data collection and annotation for other research teams, providing valuable resources that support the field of seafloor debris detection and accelerating the development and application of related technologies (Majchrowska, 2022; Deng et al., 2021).

Previous studies on seabed debris detection (Politikos et al., 2021; Xue et al., 2021; Corrigan et al., 2023) often pretrain models on the COCO or Pascal VOC datasets. While these datasets facilitate effective model transfer and reduce training time, they have inherent limitations. The objects in these datasets, which are commonly found in everyday life and captured under natural lighting conditions, differ significantly from underwater scenes. Moreover, many objects, such as cars or terrestrial wildlife, are irrelevant to the seafloor environment, leading to poor generalization in underwater scenarios. Building on these foundations and to enhance the accuracy as well as generalization of this study, TC dataset was used as a pre-training dataset of object detection models. Table 5 shows the results of various detection models with pretraining and without pretraining on the TC and COCO dataset. Most pretrained models improved significantly, except YOLOv5Lite-c. As a lightweight version, its limited capacity may have been exceeded by complex pretraining features, leading to decreased performance. The performance of another lightweight model, YOLOv3-tiny, declined in precision, likely due to the same reason. Compared to transfer learning on the COCO dataset, models pretrained on the TC dataset demonstrated better generalization and adaptability, resulting in improved feature representation. Overall, the robustness of detection models in challenging underwater conditions is significantly enhanced thereby better adapting to the seafloor debris detection. This adaptability ensures that models can be effectively utilized across a range of underwater pollution scenarios such as floating debris in rivers (Gomez et al., 2022; Niu et al., 2019); other areas of deep-sea debris (Xue et al., 2021; Bajaj et al., 2021; Zhou et al., 2022), enhancing their utility in diverse conservation contexts. Consequently, the approach not only advances the field of seafloor debris detection but also provides a versatile framework that can be extended to address broader ecological challenges in underwater environments.

#### 4.4. Quantitative analysis of seafloor debris

Seafloor debris has become a pressing global issue, adversely impacting marine ecosystems worldwide. The lack of public awareness, combined with inadequate solid waste management practices, has led to the direct discharge of substantial amounts of untreated solid waste into marine environments. Fig. 7 illustrates the distribution and proportions

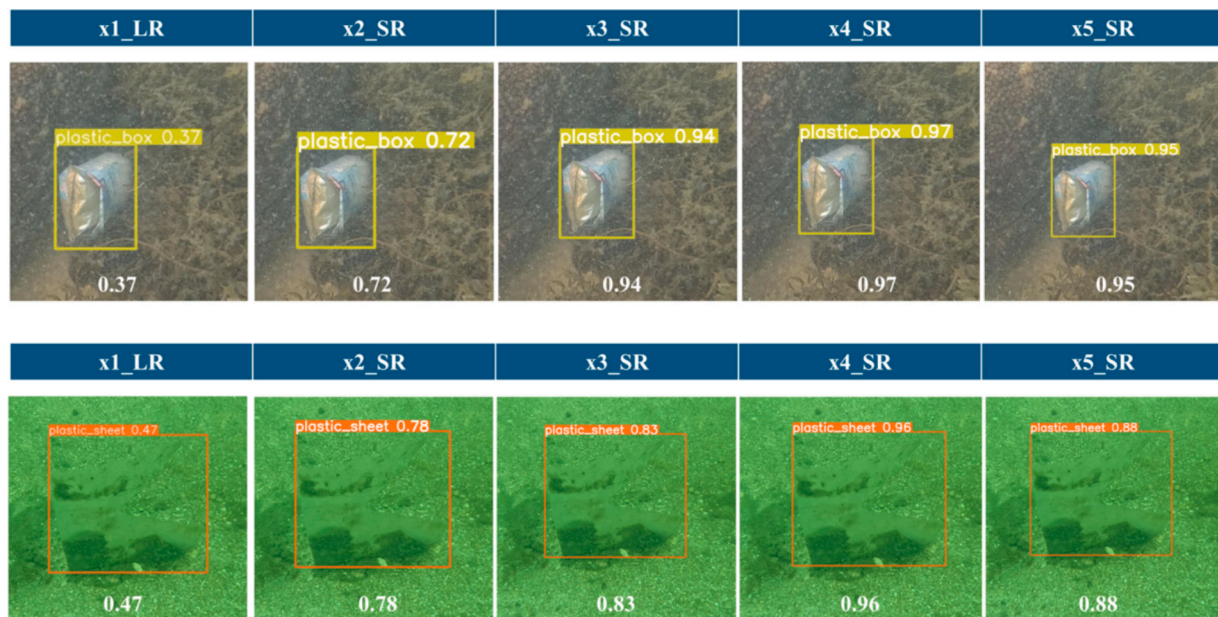


Fig. 6. Detection results of images from SR test sets obtained with different magnification factors.

**Table 5**

Evaluation metrics of mAP@0.5 (%) of detection models on different test set, HR test sets-based models categorized by pretraining method: using TrashCan, COCO and no pre training.

| Model           | Bicubic | SR (RDN) | HR (TC for pretraining) | HR (COCO for pretraining) | HR (No pretraining) |
|-----------------|---------|----------|-------------------------|---------------------------|---------------------|
| MobileNetV2-SSD | 79.5    | 89.3     | 89.6                    | 59.1                      | 19.2                |
| Faster-RCNN     | 53.2    | 60.3     | 63.3                    | 63.3                      | 27.6                |
| YOLOX-tiny      | 73.4    | 83.5     | 84.4                    | 61.8                      | 23.4                |
| YOLOv3tiny      | 79.2    | 87.0     | 89.3                    | 62.8                      | 86.0                |
| YOLOv4          | 67.6    | 87.1     | 87.6                    | 61.9                      | 18.3                |
| YOLOv5Lite-c    | 55.4    | 49.4     | 50.7                    | 61.9                      | 59.0                |
| YOLOv5s         | 76.0    | 80.6     | 81.1                    | 63.9                      | 53.5                |
| YOLOv8s         | 74.0    | 82.1     | 84.5                    | 78.4                      | 75.0                |
| SFD-YOLO        | 80.5    | 89.7     | 91.2                    | 84.7                      | 81.7                |

of seafloor debris across 7 regions (Goodman et al., 2020; Politikos et al., 2021; Loulad et al., 2019; Kuroda et al., 2020; Urban-Malinga et al., 2018; Strafella et al., 2015; Öztekin and Bat, 2017) and Koh Tao, Thailand in this paper. The figure highlights the growing prevalence of plastic waste, such as plastic bags and bottles, as these products become more widespread. Plastic debris now constitutes a significant portion of seafloor debris. For instance, plastic accounts for 73.57 % of the debris in the Moroccan Mediterranean Sea, 66.5 % in the Southern Baltic Sea, and an overwhelming 95.35 % in Sinop İnceburun Coast, Turkey. This dominance of plastic waste is a major concern for marine biodiversity.

Some marine areas, due to their proximity to industrial sites (e.g. shipyards and marinas), experience pollution from cables, metal scraps, and tires, having severely impacts on the marine environment (Goodman et al., 2020; Politikos et al., 2021). For example, in the Bay of Fundy, Canada, fishing gear including plastic-based materials, makes up 28 % of seafloor litter, with other industrial waste such as cables, metals, and tires accounting for an additional 21 %. Similarly, in Hidaka Bay, Japan, plastic debris constitutes 46 %, followed by string at 23 %, further emphasizing the impact of maritime activities. Moreover, commercial fishing activities and commercial exchanges with other countries have profoundly contributed to the debris burden in the regions like the Moroccan Mediterranean seafloor, where plastic, rubber (12.42 %), and textile (8.39 %) waste are prevalent (Loulad et al., 2019). To solve

these problems, various international conventions and programs (e.g. MARPOL and the London and Barcelona conventions) have been established to enhance debris monitoring techniques and mitigate the negative impacts on the marine environment (Fitzmaurice, 2023; Sarwar et al., 2021). Some regulations on limiting plastic usage and reducing illegal dumping, particularly by educating fishers are necessary. Additionally, addressing touristic activities impact on seafloor debris is equally essential. The study site in Ermoupolis Bay, Syros Island, Greece is a popular destination for both Greek and international tourists, known for its historic capital Ermoupolis, neoclassical architecture, cultural festivals, and stunning beaches. The waste composition of this region includes a high percentage of plastic bottles (24.7 %), plastic bags (16.5 %), and small plastic sheets (9.9 %). Other significant waste types include tires (9.1 %) and fishing nets (1.6 %), demonstrating the combined effects of tourism and local fishing practices. Similarly, Aow Luek Bay is a globally recognized diving destination that attracts visitors seeking pristine waters, coral reefs, and marine biodiversity. These similar resort destinations primarily generate plastic debris (63.55 %) which is often left behind by tourists or discarded improperly, highlighting the urgent need for clear and enforceable policies to regulate tourist behavior and promote sustainable waste management practices (Politikos et al., 2021).

Addressing this complex issue requires not only legal regulations and environmental protection initiatives but also advanced detection methods for seafloor debris. The application of deep learning technologies to achieve automatic detection of seafloor litter, with promising results that demonstrate its potential to support continuous and precise monitoring of marine waste. In this paper, a cutting-edge deep learning technique, referring SR and object detection was leveraged to address this issue more effectively. These advanced methods contribute to policy formulation and support the implementation of various environmental projects by providing accessible software and hardware solutions. The application of these technologies signifies a pivotal advancement in addressing seafloor litter, offering a more sustainable future through improved monitoring and preventive strategies.

#### 4.5. Addressing data imbalance and object similarity in underwater litter detection

The bar chart highlights the total number of objects per category for

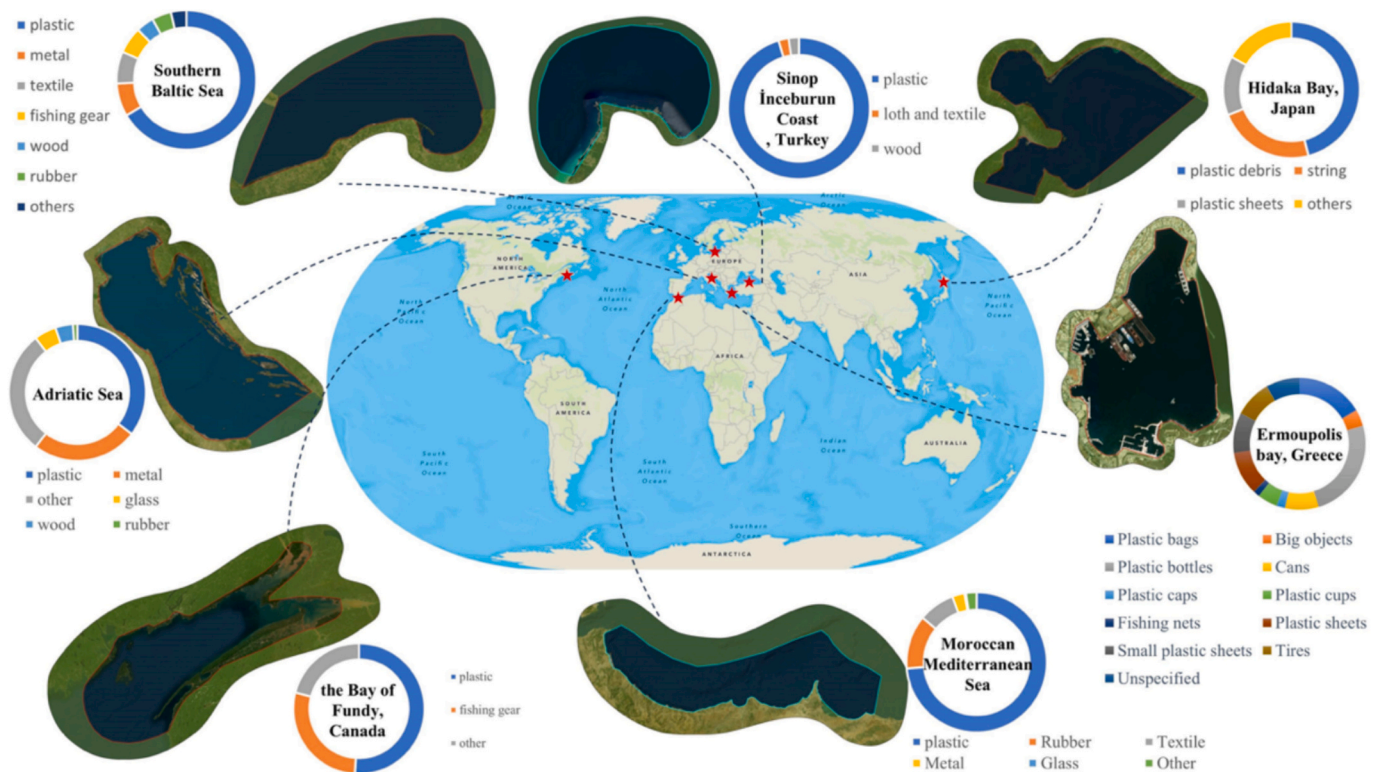


Fig. 7. Geographic distribution of studies to monitor and detect marine debris.

seafloor debris. The categories, numbered from 0 to 10, represent different types of debris such as plastic bags (label: 0), bottles (label: 1), sheets (label: 2), etc. A significant imbalance in the dataset is evident, with some categories heavily overrepresented while others are scarcely represented. For the object detection context, metal tube (label: 6) account for 30.21 % of the dataset and are the most frequent class with 116 objects. This overrepresentation may lead to the model prioritizing detection of tubes over other debris types, potentially biasing its performance. In contrast, categories such as glass bottles (label: 7), fishing nets (label: 8), ropes (label: 9), and cans (label: 10) are significantly underrepresented, with object counts ranging from 2 to 6. These account for less than 2 % of the proportion, posing challenges for the model to generalize well for these classes due to insufficient training samples. These issues of data imbalance undermine the accuracy and robustness of current models, highlighting the need for more diverse and representative training data.

To address these issues, the use of pre-training dataset Trashcan in this study can also solve imbalance of underrepresented data. This approach can enhance the model's ability to recognize plastic bags by leveraging a larger and more diverse set of training examples. Another effective solution is to use GANs (Zhu et al., 2017; Karras et al., 2018) for data augmentation by generating synthetic images for glass bottles, fishing nets, ropes, and cans. By training a GAN on real images from these classes, the model learns to generate high-fidelity synthetic images that mimic their visual characteristics. This approach increases the diversity and quantity of training data, mitigating data imbalance and enhancing model generalization in seafloor debris detection (Wang et al., 2021b).

## 5. Conclusion

This study proposes an automatic seafloor debris detection method by leveraging deep learning-based SRR and object detection techniques. The method effectively addresses the challenges posed by motion blur, murky water quality, and varying light intensity during image

acquisition with underwater cameras, which often result in low-quality images. Firstly, one Bicubic-based and six deep learning-based SRR algorithms are employed to reconstruct LR images into high-quality SR images and their performance were evaluated through the metrics of PSNR and SSIM. RDN was selected as the best SRR method due to its outstanding reconstruction ability by achieving 41.02 dB PSNR and 95.08 % SSIM. Secondly, a novel, robust, highly transferability seafloor debris object detection model, named SFD-YOLO was proposed, incorporating four innovative modules into original YOLOv8. Pretrained on TC, SFD-YOLO achieved comparable detection accuracy on HR images and RDN-generated SR images, with 91.2 % mAP and 89.7 % mAP, respectively, at the optimal magnification of 4×. Finally, its performance was compared with eight mainstream object detection models, and SFD-YOLO consistently outperformed them across Bicubic, HR, and SR test sets, making it the SOTA model in this study. However, the proposed seafloor debris detection method is divided into two separate steps—SRR and debris detection—rendering it unsuitable for real-time detection applications. Future studies will investigate the development of an optimized network that integrates SRR and object detection into an end-to-end framework. Additionally, efforts will focus on expanding the seafloor debris dataset and increasing the complexity of underwater scenes to enable the recognition and accurate identification of a broader range of debris types and conditions, ultimately leading to more reliable and comprehensive aquatic environmental surveys.

## CRedit authorship contribution statement

**Fan Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Baoxi Huang:** Writing – original draft, Visualization, Validation, Software, Methodology. **Jiaqi Wang:** Writing – review & editing, Supervision, Software, Resources, Data curation. **Xinlei Shao:** Writing – review & editing, Supervision, Resources, Investigation, Formal analysis, Data curation. **Qingyang Wu:** Writing –

review & editing, Supervision, Methodology. **Dianhan Xi:** Writing – review & editing, Visualization, Software. **Yongying Liu:** Writing – review & editing, Software, Methodology. **Yijia Chen:** Writing – review & editing, Visualization, Resources, Methodology. **Guochen Zhang:** Writing – review & editing, Validation, Methodology. **Zhiyan Ren:** Writing – review & editing, Visualization, Methodology. **Jundong Chen:** Writing – review & editing, Supervision, Methodology. **Katsunori Mizuno:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

### Author's contributions

F.Z. conceptualized the research plan, developed the data processing flow, and analyzed the experimental data; F.Z., J.W. and X. S. conducted the field survey; F.Z. and K.M. proposed the idea for super-resolution and detection of targets; F.Z., B.H. and D. X. created the training data for deep learning and visualized the results; B.H. and F.Z. labeled the data for the training of detection and made the figures; F.Z. and B.H. implemented the super-resolution and detection method, analyzed the data; K.M., X. S., Q. W., Y. L., Y. C., G. Z., Z. R. and J.C. revised the manuscript; All authors discussed the results; F.Z. and B.H. wrote the entire manuscript.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Acknowledgements

The authors extend their gratitude to the New Heaven Reef Conservation Program for their invaluable support in Thailand. Special thanks go to Mr. Jian Song and Mr. Mowen Zhang from the Graduate School of Frontier Sciences at The University of Tokyo, as well as Miss Yulun Chen from the Department of Environmental Science at Southwest Forestry University, for their significant contributions to our scientific research activities and manuscript revisions. This research was partially funded by the Japan Science and Technology Agency's SPRING Program (JST SPRING), Grant Number JPMJSP2108, and also received funding from JST, PRESTO Grant Number JPMJPR24G9, Japan.

### Data availability

The source code and dataset has been uploaded to <https://doi.org/10.5281/zenodo.14866178>. The dataset will be continuously updated to ensure its completeness and improvement.

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